T04 Permissions: Glob Patterns & Whitelist

Technical Intermediate

Precisely controlling accessible tools with glob patterns

Two pattern formats

Simple format: matches a tool name or a Bash command.

```
"permissions": {
  "allow": ["Bash(git *)", "Bash(pnpm *)", "Read", "Edit"],
  "deny":  ["Bash(rm -rf *)", "Bash(sudo *)", "Bash(npm publish)"]
}
```

Path patterns: Read, Edit and Write accept the path pattern directly. Do not add `file_path:`; that qualifier is invalid for these tools.

```
"permissions": {
  "deny": [
    "Read(*.env*)",
    "Edit(*.pem)",
    "Write(*.key)",
    "Read(*credentials*)"
  ]
}
```

Glob syntax

Pattern	What matches
`Bash(git *)`	Any git command
`Bash(npm test*)`	`npm test`, `npm test:watch`
`Read(*.env*)`	`.env`, `.env.local`, `.env.prod`
`Edit(./src/**)`	All files under `src/`
`Write(*.key)`	Any `.key` file

Progressive levels

```
// Level 1 - Auto-approve read tools
{ "permissions": { "allow": ["Read", "Grep", "Glob"] } }

// Level 2 - Git + package manager
{ "permissions": { "allow": ["Read", "Grep", "Glob",
    "Bash(git *)", "Bash(pnpm *)"] } }

// Level 3 - Full development
{ "permissions": { "allow": ["Read", "Grep", "Glob",
    "Edit", "Write", "Bash(git *)", "Bash(pnpm *)"] } }
```

Behavior of the three categories

Category	Behavior
`allow`	Auto-approved, no prompt
`deny`	Completely blocked
`ask`	Confirmation requested
(default)	Default permission mode

Known limitation

A path rule restricts matching tool access; it cannot remove a secret already pasted into a prompt or returned by another tool. Keep sensitive data outside the process access boundary and review shell, MCP and sandbox controls together.

Where to define permissions

```
~/.claude/settings.json       (global, all projects)
.claude/settings.json         (project, shared with team)
.claude/settings.local.json   (local machine, gitignore)
```

Rule evaluation: deny rules from any scope take precedence over ask and allow rules. A local allow cannot override a managed or user-level deny. Use local settings for project-specific preferences within those limits.